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Chapter 20

RIP and OSPF

In this chapter, you are presented with questions related to two industry standard routing protocols: the Routing Information Protocol (RIP) and the Open Shortest Path First (OSPF). The ICND2 exam tests you heavily on OSPF, and you may come across some RIP questions, too, so be sure to know both protocols well!

The Problems You'll Work On

In this chapter, you'll review questions concerning the following topics:

- Understanding RIP
- Configuring RIP
- Understanding OSPF
- Configuring OSPF

What to Watch Out For

Don't let common mistakes trip you up; watch for the following when working with these questions:

- RIP is a distance vector routing protocol that has two versions: RIPv1 and RIPv2. RIP has an administrative distance of 120 by default that sends routing updates every 30 seconds.
- RIPv2 is a classless routing protocol that supports VLSM and authentication.
- OSPF is a link state routing protocol that has a default administrative distance of 110. OSPF identifies neighboring routers by sending Hello messages every ten seconds.
- OSPF does not auto summarize routes by default: You must configure summarization. OSPF uses a metric know as cost, which is calculated based on the bandwidth of the link.
- OSPF minimizes bandwidth usage by electing a Designated Router (DR) that all routers will exchange information with. The DR will then send routing information to all other routers. The DR is elected by the router with the highest OSPF priority. If there are multiple routers with the same priority, then the router with the highest router ID is used. If the router ID has not been set, then a router ID is chosen by (a) the highest IP address on a loopback interface, or (b) if there are no loopback addresses, then the highest IP on a physical interface.

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Understanding RIP

814–821 Choose the best answer(s).

814. What is the administrative distance of RIP?

(A) 90

(B) 120

(C) 110

(D) 1

815. After configuring RIP, you are not receiving any RIP entries in your routing table. What command could you use to verify you are receiving RIP updates?

(A) `show rip messages`

(B) `show rip`

(C) `debug ip rip`

(D) `ip rip debug`

816. Which RIP version supports VLSM?

(A) Version 1

(B) Version 2

(C) Version 3

(D) Version 4

817. Which of the following identifies differences between RIPv1 and RIPv2? (Select two.)

(A) RIPv2 sends updates every 60 seconds.

(B) RIPv2 is classful.

(C) RIPv2 sends updates using multicast communication.

(D) RIPv2 sends updates using broadcast communication.

(E) RIPv2 supports authentication.

818. You have configured RIPv2 and you do not want it to auto summarize routes. What command would you use?

(A) `delete auto-summary`

(B) `no auto-summary`

(C) `negate summary`

(D) `auto-summary stop`

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819. Looking at the figure below, what version of RIP has been enabled?

```
R1#debug ip rip
RIP protocol debugging is on
R1#RIP: received v1 update from 12.0.0.2 on Serial0/3/0
      13.0.0.0 in 1 hops
RIP: sending  v1 update to 255.255.255.255 via FastEthernet0/0 (11.0.0.1)
RIP: build update entries
      network 12.0.0.0 metric 1
      network 13.0.0.0 metric 2
RIP: sending  v1 update to 255.255.255.255 via Serial0/3/0 (12.0.0.1)
RIP: build update entries
      network 11.0.0.0 metric 1
```

(A) Version 2

(B) Version 1

(C) Version 4

(D) Version 3

820. How frequently does RIPv2 send out routing table updates?

(A) Every 30 seconds

(B) Every 60 seconds

(C) Every 90 seconds

(D) Every 120 seconds

821. You want to change the administrative distance of RIP to a value other than its default. What command would you use?

(A) R1(config-router)#`distance 80`

(B) R1(config-router)#`distance 120`

(C) R1(config)#`distance 80`

(D) R1(config)#`distance 120`

Configuring RIP

822–829 Choose the best answer(s).

822. You have already configured RIPv1, but have recently learned of the benefits of RIPv2. What additional command(s) would you use to enable RIPv2?

(A) R1(config-router)#`version 2`

(B) R1(config-router)#`enable version 2`

(C) R1(config)#`version 2`

(D) R1(config-router)#`no version 1`

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823. Looking at the figure below, what is the administrative distance?

```
R1#show ip protocols
Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 11 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip
Default version control: send version 1, receive any version
  Interface             Send  Recv  Triggered RIP  Key-chain
  FastEthernet0/0       1     2 1
  Serial0/3/0           1     2 1
Automatic network summarization is in effect
Maximum path: 4
Routing for Networks:
        11.0.0.0
        12.0.0.0
Passive Interface(s):
Routing Information Sources:
        Gateway         Distance      Last Update
        12.0.0.2              25      00:00:16
Distance: 25 (default is 120)
```

(A) 25

(B) 30

(C) 180

(D) 120

824. You have configured RIP on your router and want to view the entries in the routing table. What command would you use?

(A) `show rip`

(B) `show rip entries`

(C) `show ip route`

(D) `show ip rip`

825. You have a router with two networks you wish to advertise through RIP – the 192.168.3.0 and the 192.168.4.0 networks. Which of the following identifies the commands you would use?

(A) Use the following commands:

R1(config)#`router rip`

R1(config-router)#`network 192.168.3.0`

R1(config-router)#`network 192.168.4.0`

(B) Use the following commands:

R1(config)#`network 192.168.3.0`

R1(config)#`network 192.168.4.0`

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(C) Use the following commands:

R1(config)#`router rip`

R1(config-router)#`network 192.168.3.0 0.0.0.255`

R1(config-router)#`network 192.168.4.0 0.0.0.255`

(D) Use the following commands:

R1(config)#`router rip`

R1(config-router)#`network 192.168.3.0 area 1`

R1(config-router)#`network 192.168.4.0 area 1`

826. Looking at the figure below, how many hops away is the 14.56.7.8 system?

```
R2#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

R    11.0.0.0/8 [120/1] via 12.0.0.1, 00:00:05, Serial0/3/0
R    14.0.0.0/8 [120/3] via 12.0.0.5, 00:00:05, Serial0/3/0
C    12.0.0.0/8 is directly connected, Serial0/3/0
C    13.0.0.0/8 is directly connected, FastEthernet0/0
R2#
```

(A) 1

(B) 120

(C) 5

(D) 3

827. You want to view how frequently RIP is sending updates. What command would you use?

(A) `show rip`

(B) `show protocols`

(C) `show rip config`

(D) `show ip protocols`

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828. Looking at the figure below, what command was used to create the output?

```
Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 16 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip
Default version control: send version 1, receive any version
  Interface             Send  Recv  Triggered RIP  Key-chain
  Serial0/3/0           1     2 1
  FastEthernet0/0       1     2 1
Automatic network summarization is in effect
Maximum path: 4
Routing for Networks:
        12.0.0.0
        13.0.0.0
Passive Interface(s):
Routing Information Sources:
        Gateway         Distance      Last Update
        12.0.0.1             120      00:00:20
Distance: (default is 120)
R2#
```

(A) `show ip route`

(B) `show ip protocols`

(C) `show rip`

(D) `show ospf`

829. Which of the following commands would you use to change the administrative distance with RIP?

(A) Use the following commands:

R1(config)#`router rip`

R1(config-router)#`distance 44`

(B) Use the following commands:

R1(config)#`router eigrp`

R1(config-router)#`distance 44`

(C) Use the following commands:

R1#`router rip`

R1#`distance 44`

(D) Use the following command:

R1(config)#`distance 44`

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Understanding OSPF

830–842 Choose the best answer(s).

830. What is the maximum number of equal cost routes that can exist in the routing table with OSPF?

(A) 1
(B) 2
(C) 4
(D) 8

831. What is the default administrative distance of routes learned with OSPF?

(A) 90
(B) 120
(C) 110
(D) 1

832. Which of the following does OSPF use as a metric value?

(A) Bandwidth
(B) Hop count
(C) Cost
(D) Distance

833. When the OSPF priorities on all the routers are the same, how is the DR selected?

(A) The router with the lowest router ID
(B) The router with the highest router ID
(C) The router with the priority set to 0
(D) One is randomly selected.

834. How frequently are hello messages sent with OSPF?

(A) Every 90 seconds
(B) Every 60 seconds
(C) Every 30 seconds
(D) Every 10 seconds

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835. Which of the following areas is considered the "backbone" area with OSPF?

(A) Area 1

(B) Area 0

(C) Area 2

(D) Area 10

836. How frequently does RIPv1 send out routing table updates?

(A) Every 30 seconds

(B) Every 60 seconds

(C) Every 90 seconds

(D) Every 120 seconds

837. What is the purpose of a Designated Router (DR) with OSPF?

(A) It acts as a backup if the BDR fails.

(B) It assigns IP addresses to clients on the network.

(C) It converts private addresses to public addresses.

(D) All other routers exchange info with the DR to cut down on bandwidth usage.

838. You wish to ensure that router R1 never becomes the DR. What should you do?

(A) Set the priority to 0.

(B) Set the priority to 1.

(C) Set the priority to 99.

(D) Set the priority to 255.

839. How often are hello messages sent with OSPF?

(A) Every 12 seconds

(B) Every 20 seconds

(C) Every 10 seconds

(D) Every 30 seconds

840. What happens when the DR fails?

(A) The BDR becomes the DR and an election for a new BDR occurs.

(B) A new DR is elected.

(C) The router ID is changed.

(D) The priority of the OSPF router is set to 0.

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841. What are two reasons why two OSPF routers would not be able to create neighbor relationships? (Select two.)

(A) The router IDs are different.

(B) Hello and Dead Interval timers are not configured the same on both routers.

(C) The routers are in the same area.

(D) The routers are in different areas.

(E) There is no loopback interface configured on each router.

842. What is the purpose of the BDR with OSPF?

(A) The BDR informs routers of priorities.

(B) The BDR contains fault tolerant links to head office.

(C) The BDR contains fault tolerant links to a central network.

(D) The BDR is used when the DR fails.

Configuring OSPF

843–851 Choose the best answer(s).

843. Looking at the commands below, what is the purpose of the "1"?

```
Router ospf 1
Network 192.168.2.0 0.0.0.255 area 0
```

(A) It is the OSPF priority.

(B) It is the router ID.

(C) It is the process ID for OSPF.

(D) It is a bit flag setting OSPF to "on."

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844. Looking at the figure below, what does the /65 indicate in the third route of the routing table?

```
R1>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    11.0.0.0/8 is directly connected, FastEthernet0/0
C    12.0.0.0/8 is directly connected, Serial0/3/0
O    13.0.0.0/8 [110/65] via 12.0.0.2, 00:42:33, Serial0/3/0
R1>
```

(A) The administrative distance

(B) The bandwidth

(C) The hop count

(D) The cost

845. Your router has the interface configuration shown below. What is the router ID of the router?

Loopback0: 12.0.0.1

Loopback1: 14.0.0.1

F0/1: 24.0.0.1

F0/2: 96.0.0.1

(A) 14.0.0.1

(B) 12.0.0.1

(C) 24.0.0.1

(D) 96.0.0.1

846. Looking at the figure below, what would the router-ID of the router be?

```
R2#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        13.0.0.1        YES manual up                    up
FastEthernet0/1        unassigned      YES unset  administratively down down
Serial0/3/0            12.0.0.2        YES manual up                    up
Serial0/3/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R2#
```

(A) 12.0.0.1

(B) R1

(C) R2

(D) 13.0.0.1

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847. What commands would you use to configure OSPF and add network 192.168.5.0/24 to area 0? (Select two.)

(A) `router ospf area 0`

(B) `router ospf 1`

(C) `network 192.168.5.0 0.0.0.255 area 0`

(D) `network 192.168.5.0 255.255.255.0`

(E) `network 192.168.5.0 0.0.0.255`

848. You want to change the OSPF priority of router R1. What command would you use?

(A) `ospf priority 20`

(B) `ip priority ospf 20`

(C) `priority ospf 20`

(D) `ip ospf priority 20`

849. Which of the following is used to calculate OSPF cost?

(A) Bandwidth

(B) Delay

(C) Hop count

(D) Latency

850. You want to change the router ID of a Cisco router running OSPF. What command would you use?

(A) R1(config-router)#`router-id 105`

(B) R1(config-router)#`router-id 13.0.0.1`

(C) R1(config)#`router-id 13.0.0.1`

(D) R1#`router-id 105`

851. Hello messages are sent to which of the following addresses?

(A) 255.255.255.255

(B) 224.0.0.5

(C) 239.0.0.5

(D) 255.0.0.0